

CAFE: Concept-Anchored FacEts for Enabling Domain-Grounded Natural Language Explanations (Supplementary Material)

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I. MORE RELATED WORK

In this section, we extend our literature review to explainability in Legal and Medical AI.

A. Explainability in Legal AI

[1] employs BERT-based transformers followed by SVM and XGBoost classifiers for bail prediction and salient sentence recognition, respectively. A multitask learning framework is utilized, where improved salient sentence detection contributes to more accurate predictions. This approach also enables the salient sentences to serve as explanations. [2] adopts a hierarchical XLNet-based architecture with BiGRU units for context regularization in the judgment prediction task. An occlusion-based method, inspired by [3] and [4], is applied wherein, for each document processed by the BiGRU component, each complete chunk of the embedding is masked individually, and the resulting output probability is compared against the unmasked output. A positive difference indicates that the masked chunk contributes to the model’s decision and is thus considered explanatory. [5] learns a mask over case facts, compressing the representation to retain only the relevant facts, and performs alleged article violation prediction. While these methods address explainability, they do not incorporate legal precedents into their prediction processes. [6] introduces a precedent-enhanced approach that leverages domain models to identify similar cases—referred to as precedents—from a case database and employs Retrieval-Augmented Generation (RAG) [7] with a large language model (LLM) to improve judgment prediction. Although this approach incorporates precedents, LLMs inherently lack explainability, and precedents can only serve as explanations after manual filtering of their factual content. Moreover, the approach requires the case database to conform to a specific format, limiting its applicability to real-world, unstandardized data. [8] utilizes both attribute and structural encoders to learn case embeddings. The structural encoder captures a citation network organized as a hierarchical graph, where nodes represent statutes, their chapters, topics, and sections, with links indicating hierarchical relationships. Case facts (i.e., portions of a case) are also represented as nodes and connected to sections they cite. Inter- and intra-metapath aggregation are applied to derive the structural encoding, while a Hierarchical Attention Network [9] is used to jointly encode the attributes of facts and sections. Formulated as an inductive link prediction task between a

newly introduced fact and a section, this Legal Statute Identification method focuses on individual case facts rather than entire cases and does not provide explainability.

B. Explainability in Medical AI

[10] provides a multidisciplinary assessment of explainability in medical AI, emphasizing that it extends beyond technical transparency to encompass ethical and legal accountability in clinical decision support. While they note that post-hoc techniques such as LIME and SHAP provide interpretability, these methods remain detached from the structural logic of clinical reasoning and fail to capture how individual medical elements interact. Our proposed Bifacet Graph framework addresses this by embedding medical cases within a hierarchical structure that encodes the relationships among findings, diagnoses, and procedures. This allows explanations to emerge as subgraphs that reflect clinical dependencies, maintaining transparency while preserving contextual meaning. [11] conducts a comprehensive review of explainable and interpretable deep learning methods in healthcare NLP, identifying attention mechanisms and knowledge-graph-based reasoning as dominant strategies. However, most of these approaches operate at a local level and cannot generalize to global, system-level reasoning across complex medical data. By training a BiGNN for transductive link prediction over the Bifacet Graph, our framework extends explainability beyond local feature importance to capture relational patterns between medical entities, enabling global structural interpretability while retaining post-hoc flexibility. [12] presents a meta-analysis of explainable AI in clinical decision support systems, highlighting the prevalence of model-agnostic methods such as Grad-CAM, SHAP, and counterfactuals across domains like radiology and oncology. They point out that current XAI techniques struggle with fragmented data representation, low explanation fidelity, and limited usability in clinical workflows. In contrast, our method provides a unified, graph-based representation of the medical case database, from which subgraph explanations are extracted and translated into natural language using an LLM. This produces faithful, relationally grounded, and human-readable explanations that integrate seamlessly with clinical reasoning, overcoming the opacity and fragmentation inherent in existing XAI methods.

Component Type	Count
Nodes	
Articles	116
Terms	440
Facts	212,878
Cases	9,000
Edges	
Articles $\leftrightarrow$ Terms	1,565
Facts $\leftrightarrow$ Terms	1,081,399
Cases $\leftrightarrow$ Articles	12,115
Facts $\leftrightarrow$ Cases	212,878

TABLE I: Summary of Bifacet LegalGraph Structure Statistics

II. CONSTRUCTION OF BIFACET GRAPHS

In this section, we describe the legal and medical datasets used to construct the respective Bifacet Graphs, the construction process, and the tasks for which the BiGNNs were trained.

A. Legal Domain – Bifacet LegalGraph

The Bifacet LegalGraph is constructed using the European Court of Human Rights (ECtHR) dataset [5]. This dataset comprises 11,000 legal cases, each entry providing structured information including detailed facts, alleged and adjudicated violations of the European Convention on Human Rights (ECHR), and expert- or court-identified rationales (termed silver or gold rationales) that justify the judgments. This rich structure makes it ideal for tasks like legal judgment prediction, rationale extraction, and prior case retrieval.

The knowledge code nodes represent the allegedly violated ECtHR articles and the fundamental nodes comprise law-specific terms tokenized from the article descriptions. The document nodes include level 1 nodes for individual paragraphs (facts) and level 2 nodes for the complete legal case texts. All node attributes leverage LegalBERT embeddings [13]. Specifically, $\mathcal{F}$ nodes are embedded as the legal term, $\mathcal{K}$ nodes as the full article description, and $\mathcal{D}_1$ nodes as the individual paragraph text. For the high-level $\mathcal{D}_2$ case nodes, where concatenated paragraphs often exceed LegalBERT’s context limit, a sliding window is used: the final attribute is the mean of the LegalBERT encodings across all resulting windows. The graph structure statistics are summarized in Table I.

The structure of the Bifacet LegalGraph naturally supports two core tasks in our evaluation. First, Transductive Edge Prediction: predicting links between legal cases $\mathcal{D}_2$ and ECtHR articles $\mathcal{K}$, corresponding to the Legal Judgment Prediction problem. Second, Explanation Generation: for each positively predicted case–article link, we produce a justification via Rationale Extraction, drawing on relevant precedential case pathways in the graph. The latter implicitly addresses the Prior Case Retrieval problem by identifying relevant evidential paths in the graph.

B. Medical Domain – Bifacet MedGraph

For the medical domain, we construct the Bifacet MedGraph using the MIMIC-IV dataset [14]. This comprehensive dataset

contains de-identified electronic health records (EHRs) from the Beth Israel Deaconess Medical Center to support the task of Medical Procedure Necessity Prediction. In this task, we predict the necessity of specific medical procedures (ICD-10-PCS codes [15]) based on the clinical context provided in discharge summaries, i.e., Medical Procedure Necessity Prediction (MPNP). In clinical practice, determining the necessity of a procedure requires a complex reasoning process that maps a patient’s unique clinical presentation to standardized medical protocols. Our preprocessing pipeline is designed to transform the raw, complex data into the Bifacet Graph structure, linking clinical evidence to procedural knowledge. This pipeline involves three primary stages: Knowledge Code (ICD) node generation, Document (Discharge Summary) node generation, and Foundation (Medical Entity) node extraction.

The knowledge codes ($\mathcal{K}$) in the Bifacet MedGraph are represented by ICD-10 procedure codes. To build rich text attributes for these nodes, we initiate a two-stage process. First, we extract the official titles and descriptions for each `icd_code` from the standard ICD-10-PCS code files published by the Centers for Medicare & Medicaid Services (CMS) [15]. We then employ a medical language model (Clinical T5 [16], trained on MIMIC-III [17] and IV with ICD codes) to enrich these formal definitions, transforming the compressed titles into multi-sentence descriptions that serve as the primary text attribute for each knowledge code node $k \in \mathcal{K}$.

The document axis of the graph is constructed from the MIMIC-IV discharge summaries. Each summary, tied to a specific `subject_id` and hospital admission (`hadm_id`), represents a high-level Level 2 ($\mathcal{D}_{L=2}$) document node. To create fine-grained Level 1 ($\mathcal{D}_1$) nodes, we parse the unstructured `text` field of each discharge summary using a regular expression-based method to identify and extract free-text content following a predefined set of section headers: Chief Complaint, Major Surgical or Invasive Procedure, History of Present Illness, Past Medical History, Social History, Family History, Physical Exam, Brief Hospital Course, Pertinent Results, Medications on Admission, Discharge Medications, Discharge Instructions, Discharge Diagnosis. These individual text sections function as Level 1 nodes, hierarchically linked to their parent Level 2 admission node. Ground-truth edges ($\mathcal{E}_{DK}$) are established by merging the parsed document data with the MIMIC-IV admission-to-code files, connecting each `hadm_id` to its corresponding ICD procedure codes and forming the observed edge set $\hat{\mathcal{E}}_{DK}$.

To extract the shared medical terminology bridging the knowledge and document axes, we perform Named Entity Recognition (NER) using the Clinical-AI-Apollo/Medical-NER model [18] on both ICD code descriptions and parsed clinical notes. We filter for entities belonging to the following semantic groups: BIOLOGICAL_STRUCTURE,

Component Type	Count
Nodes	
Notes	42,755
ICDs	8,367
Terms	5,759
Edges	
Notes $\leftrightarrow$ Terms	2,137,283
Notes $\leftrightarrow$ ICDs	125,653
ICDs $\leftrightarrow$ Terms	44,561

TABLE II: Summary of Bifacet MedGraph Statistics

DIAGNOSTIC_PROCEDURE, DISEASE_DISORDER, HISTORY, MEDICATION, SEVERITY, SIGN_SYMPTOM, and THERAPEUTIC_PROCEDURE. The set of all unique entity strings forms the fundamental node set F . Graph edges are then established as follows:

- An edge in $\mathcal{E}_{FK}$ is created between a fundamental node $f \in F$ (e.g., “hydrocephalus”) and a knowledge code $k \in \mathcal{K}$ if f is extracted from k ’s detailed description.
- An edge in $\mathcal{E}_{FD}$ is created between a fundamental node $f \in F$ (e.g., “ascites”) and a Level 1 document node $d \in D_l$ if f is extracted from the text of d .

This complete pipeline results in the final Bifacet MedGraph: a heterogeneous, Bifacet Graph linking patient admissions to their associated medical procedures via a shared foundational layer of extracted medical concepts.

III. IMPLEMENTATION DETAILS

We implement 3-layer BiGNN variants, since from an input document node d , potential precedents are 2 hops away, and articles with which it shares fundamental nodes are 3 hops away. GATConv [19] layers are used for node feature aggregation and ReLU as the activations. The dot product of the learned node features of the document and knowledge code nodes is taken as the logit for link prediction. This same architecture is used for Bifacet LegalGraph, Bifacet MedGraph, HeteroLegalGraph, and HeteroMedGraph. However, only 2-layer GNNs are used for HeteroLegalGraph and HeteroMedGraph to account for similar information being aggregated across both implementations.

2-layer GNNs with GATConv and dropout are implemented for LegalGraph and MedGraph, with a fully connected layer to flatten features for the node classification task. The multilabel node classification task is reformulated as $|\mathcal{K}|$ binary node classification tasks, with Binary Cross Entropy as the loss function.

For TabularLegal, we reimplement [5], in which they implement HIERBERT-HA [20], a hierarchical version of BERT with hard attention, and propose rationale constraints such as sparsity, comprehensiveness, and singularity as regularizers. We modify the loss function slightly by using Distribution Loss [21] instead of Binary Cross Entropy to account for the dominance of negative labels (each case document in the ECtHR dataset violates 1.35 articles on average out of 40 total, with a maximum of 8). TabularMed is implemented as

a deep neural network with 10 Linear layers accompanied by LayerNorm, ReLU, and dropout, with Asymmetric Loss [22] as the loss function.

As the Graph Explainer Φ , we develop HeteroGNNExplainer, a PyTorch Geometric-compatible heterogeneous graph variant of GNNExplainer [23]. HeteroGNNExplainer is used to learn an instance-level mask for each true positive edge prediction. It is trained on each instance for 50 epochs with a learning rate of 0.01, an edge mask sparsity regularizer (α_1) of 0.005, and an edge mask entropy regularizer (α_2) of 1.0.

The beam search algorithm with $m = 2$ and $m = 3$ is applied for a comparative analysis, selecting the top m neighbours of each node type. The centers are the source document node d and the target knowledge code node k . Bidirectional Dijkstra’s algorithm is applied on the subgraph explanations to extract the top $p = 5$ paths from d to k , with all edges in a previously extracted path dropped iteratively. Finally, for the Bifacet LegalGraph we use SaulLM-7B-Instruct [24] as our LLM, and for the Bifacet MedGraph we use BioMedical LLaMA-3-8B [25] with prompt templates given in Appendix VIII.

The BiGNNs trained on the Bifacet LegalGraph and Bifacet MedGraph, and their non-hierarchical variants HeteroLegalGraph and HeteroMedGraph, were trained with the Adam optimizer [26] at a learning rate of 5×10^{-4} for 500 epochs. The GNN trained on LegalGraph used a learning rate of 1×10^{-3} for 100 epochs. TabularLegal was trained for 100 epochs with batch size 64 and learning rate 1×10^{-3} using the AdamW optimizer [27]. TabularMed was trained for 150 epochs with batch size 128, learning rate 5×10^{-4} , and L2 weight decay 1×10^{-4} using AdamW. The Asymmetric Loss used for TabularMed had $\gamma_{\text{neg}} = 4$, $\gamma_{\text{pos}} = 4$, $\text{clip} = 0.05$, and $\epsilon = 0.1$.

IV. LIKERT SCALE STUDY

For the legal domain, a qualified legal expert evaluated 20 comparative explanation sets, each comprising a baseline, silver rationale, and *BiFacetExplainer* output for a unique case–article pair. Similarly, two medical professionals assessed 11 comparative sets within the clinical domain. A representative explanation set for Case 1253 and Article 6 of the ECtHR dataset is illustrated in Figure 1.

Due to the data privacy and redistribution clauses mandated by PhysioNet, the qualitative explanation sets for MIMIC-IV are omitted from this work to maintain full regulatory compliance with the dataset’s access requirements.

V. ABLATION STUDIES

We perform an ablation study to evaluate the effectiveness of our proposed edge-selection strategy. Specifically, we compare our beam search methodology (BFE-SEARCH) against a simpler, threshold-based approach (BFE-MASK). The BFE-MASK methodology computes the subgraph explanation $\mathcal{G}_{dk}$ by applying a fixed threshold $\tau_{\text{mask}} = 0.385$ (chosen after empirical observation) to the edge weights produced by the graph explainer’s mask, dropping edges at or below this value.



Fig. 1: Explanation set from ECtHR Legal Judgement Prediction: The figure displays three explanations for a single case–article pair (Case 1253, Article 6) from the ECtHR dataset. In the top left is the baseline explanation, in the top right is the silver explanation, and in the bottom center is the explanation from our proposed *BiFacetExplainer*.

Unlike BFE-SEARCH, BFE-MASK requires manual tuning of a subgraph-specific hyperparameter and may produce highly sparse subgraphs, potentially severing all valid paths between the document node d and the knowledge code node k . To quantify this structural trade-off, we measure the average percentage of test samples retaining at least one $d \rightarrow k$ path in $\mathcal{G}_{dk}$ for both methods (with $m = 3$). The results are reported in Figure 2. The Bifacet MedGraph retains at least one valid path for 100% of samples under BFE-SEARCH, while BFE-MASK yields a lower percentage of valid paths in both domains, demonstrating that path-guided extraction produces more consistently grounded explanations.

During the construction of the Bifacet MedGraph, we observed that the same fundamental medical term often appeared multiple times within a single medical note, resulting in multiple parallel edges between a fundamental node and a knowledge code node. To assess the impact of these duplicate connections, we conducted an ablation study retaining only

a single instance of each duplicated edge, yielding a modified graph referred to as NoDupBifacetMedGraph (statistics summarized in Table III). We trained the same BiGNN on this deduplicated graph and report AUROC and F1 performance in Figure 3. GEF scores from HeteroGNNExplainer are presented in Table IV. The results indicate that although duplicate edges provide a modest improvement in AUROC and F1, the NoDupBifacetMedGraph yields explanation subgraphs with substantially higher graph faithfulness, suggesting that removing duplicate edges leads to explanations that more accurately reflect the model’s true decision-making pathways.

VI. ADDITIONAL BENEFITS OF BIFACET GRAPHS

In addition to the enriched internode relation encoding and robust precedent identification capabilities, the proposed Bifacet Graph structure offers several distinct advantages, which we detail below.

Scaling Bifacet Graphs. The Bifacet Graph is designed for efficient, dynamic updates. Adding a new knowledge

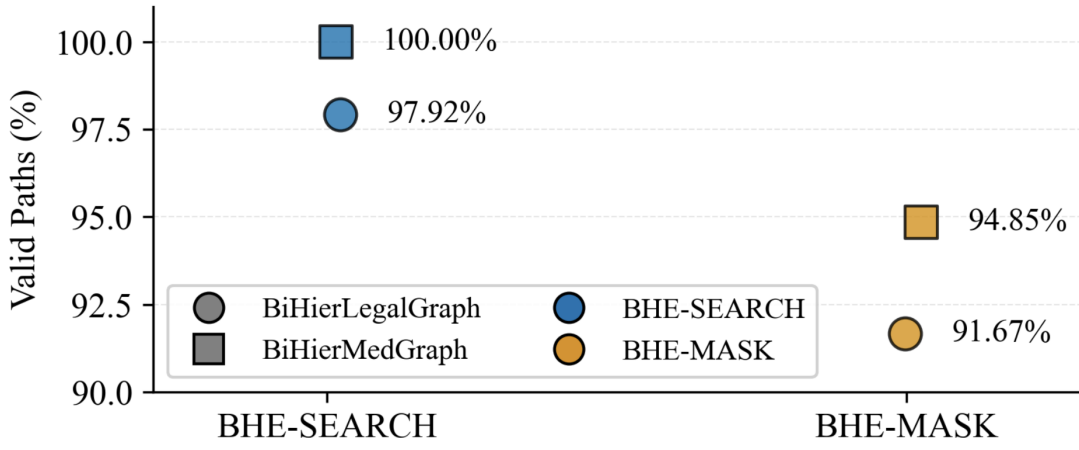


Fig. 2: **Percentage of Extracted Explanation Paths Across Bifacet Graph Domains and Methods.** The y-axis reports the percentage of (d, k) pairs that have at least one valid path from $d \rightarrow k$ in the subgraph explanation $\mathcal{G}_{dk}$, while the x-axis compares two explanation extraction strategies: BFE-SEARCH (path-based) and BFE-MASK (mask-based). Circles denote results on the Bifacet LegalGraph and squares denote results on the Bifacet MedGraph. Blue markers correspond to BFE-SEARCH and orange markers correspond to BFE-MASK.

Component Type	Count
Nodes	
Notes	42,755
ICDs	8,367
Terms	5,759
Edges	
Notes $\leftrightarrow$ Terms	1,735,382
Notes $\leftrightarrow$ ICDs	125,653
ICDs $\leftrightarrow$ Terms	44,559

TABLE III: Summary of NoDupBifacetMedGraph Statistics

Explanation	neighbours	GEF
Bifacet MedGraph	m = 2	0.4037
	m = 3	0.3870
NoDupBifacetMedGraph	m = 2	0.3734
	m = 3	0.3090

TABLE IV: Comparison of explanation faithfulness (GEF) for HeteroGNNExplainer subgraphs from Bifacet MedGraph and NoDupBifacetMedGraph, evaluated across beam search neighbour configurations m .

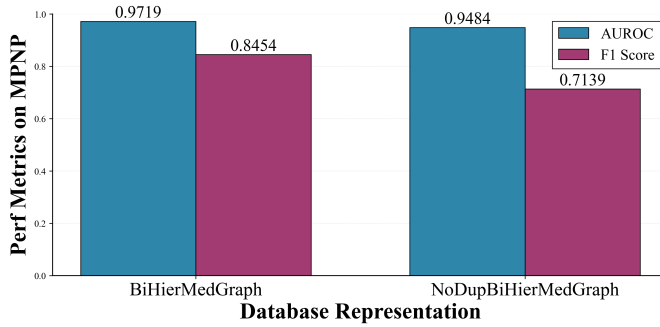


Fig. 3: **Performance comparison across dataset representations.** Bar plots showing predictive performance of Bifacet MedGraph and NoDupBifacetMedGraph on Medical Procedure Necessity Prediction (MIMIC-IV). Blue: AUROC; Purple: F1.

code node takes $\mathcal{O}(\mathcal{L}_k)$ time, where $\mathcal{L}_k$ is the length of the knowledge code text, as the update algorithm iterates over each token to query a hash table and establish edges with the existing fundamental nodes. When a new document node is registered, a two-step process is followed: first, a subgraph for

the document and its hierarchical levels is built; next, the same $\mathcal{O}(\mathcal{L}_d)$ algorithm connects its lowest-level nodes $d \in D_1$ to the fundamental nodes. A subsequent $\mathcal{O}(|\mathcal{K}|)$ operation adds edges between the new document node and all knowledge codes. Consequently, the overall time complexity for adding a new document node is $\mathcal{O}(\mathcal{L}_d + |\mathcal{K}|)$. Moreover, due to the l -layer architecture of the BiGNN, a node’s embedding is influenced exclusively by its l -hop neighbourhood, enabling an incremental training strategy where only newly added nodes and their l -hop subgraphs need to be updated.

Model-agnostic design. *BiFacetExplainer* is fundamentally model-agnostic. It does not depend on the internal architecture, training objective, or message-passing formulation of the underlying graph model. Instead, it operates on structural and representational outputs—such as node embeddings, edge weights, or attention maps—produced by any graph-based predictor, whether homogeneous or heterogeneous. This abstraction allows *BiFacetExplainer* to interface seamlessly with existing GNNs, relational models, or symbolic graph frameworks without requiring model retraining or modification, acting as a unifying interpretability layer across diverse graph learning approaches.

VII. LIMITATIONS

While Bifacet Graphs and *BiFacetExplainer* provide structured and interpretable reasoning, they are not without limitations. Certain aspects of graph construction, reasoning extraction, synthesis, and scalability introduce challenges that must be carefully managed.

Hierarchy construction. Bifacet Graphs require careful selection of fundamental nodes and hierarchical structures. Without domain expertise or robust extraction heuristics, the graph may misrepresent relationships or omit key concepts, which is critical for reliable explanations.

Path extraction limitations. The path extraction process may miss reasoning patterns that are non-linear or distributed, such as aggregated diffusion signals. As a result, some inferential links may remain unrepresented, limiting the completeness of generated explanations.

Residual LLM hallucinations. Deterministic extraction reduces hallucinations but cannot fully prevent them during the synthesis step. Careful prompt design, confidence scoring, and verification are still needed, and outputs must be critically evaluated for factual accuracy.

Scaling challenges. Despite the scaling benefits of Bifacet Graphs, extremely large graphs can introduce performance and storage challenges. Efficient sparse indices, candidate pruning, and optimized graph traversal are needed to maintain scalability at inference and explanation generation time.

Restriction to true positives. *BiFacetExplainer* currently applies only to true positive predictions, as existing graph explanation models are designed to operate on correctly classified instances. This limits analysis of false positives or false negatives; future work could integrate graph counterfactual explainers [28] that generalize to all prediction outcomes.

One-to-one document–knowledge code explanations. The explanation generation pipeline is designed for one-to-one document–knowledge code predictions. Generating explanations for a document associated with multiple knowledge codes requires running the pipeline separately for each pair, limiting efficiency in multi-label or dense knowledge code assignment settings.

VIII. PROMPT TEMPLATES

The prompt templates used in our experiments are illustrated in Figures 4, 5, and 6 for SaulLM [24], and Figures 7 and 8 for BioMedical-LLaMA [25].

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You're a lawyer working on a case on Human Rights. You have to generate an argument to prove that the defendant has violated a European Court of Human Rights article using the information that you're given.

The following is the case against the plaintiff:

<caseText>

(For each node in the path, preface the node text with the following template)
The following is a <nodeType> (violated/mentioned/contained) by the previously mentioned <nodeType>:

<nodeText>

The following is the article you want to argue that the defendant violated:

<articleText>

Using the cases mentioned as precedents, build an argument to prove that the defendant is guilty of violating the article just mentioned.
Citing relevant precedents using the cases I mentioned, give me only the argument in the following format: Introduction, Violation(s), Precedent(s), Conclusion.

Fig. 4: Prompt template used to generate *BiFacetExplainer* explanations with SaulLM.

You're a lawyer working on a case on Human Rights. You have to generate an argument to prove that the defendant has violated a European Court of Human Rights article using the information that you're given.

The following is the case against the plaintiff:

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The following is the article you want to argue that the defendant violated:

<articleText>

Using the cases mentioned as precedents, build an argument to prove that the defendant is guilty of violating the article just mentioned.
Citing relevant precedents using the cases I mentioned, give me only the argument in the following format: Introduction, Violation(s), Precedent(s), Conclusion.

Fig. 5: Baseline prompt template used with SaulLM.

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You're a lawyer working on a case on Human Rights. You have to generate an argument to prove that the defendant has violated a European Court of Human Rights article using the information that you're given.

The following is the case against the plaintiff:

<caseText>

The following is a list of very important facts, each mentioned inside a <sr></sr> block, of the case that you must focus on while generating an argument. (Add each silver rationale in a separate <sr></sr> block)

<sr>silverRationale</sr>

The following is the article you want to argue that the defendant violated:

<articleText>

Using the cases mentioned as precedents, build an argument to prove that the defendant is guilty of violating the article just mentioned.

Citing relevant precedents using the cases I mentioned, give me only the argument in the following format: Introduction, Violation(s), Precedent(s), Conclusion.

Fig. 6: Silver-rationale prompt template used with SaulLM.

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You are now a medical insurance agent working on a case on medical necessities. You have to generate arguments to prove that the medical procedures administered by the hospital are deemed a medical necessity using the information that you're given.

The following is the medical case of the patient:

<caseText>

(For each node in the path, preface the node text with the following template)
The following is a <nodeType> that the previous <nodeType> (required/contained):
<nodeText>

The following are the procedures you want to argue that the patient needs:

<icdText>

Build an argument to prove that the patient needed the procedure mentioned earlier.

Citing relevant parallels/precedents, give me ONLY the argument in the following format: Introduction, Procedures, Similar Cases, Conclusion.

Fig. 7: *BiFacetExplainer* prompt template used with BioMedical-LLaMA.

You are now a medical insurance agent working on a case on medical necessities. You have to generate arguments to prove that the medical procedures administered by the hospital are deemed a medical necessity using the information that you're given.

The following is the medical case of the patient:

<caseText>

The following are the procedures you want to argue that the patient needs:

<icdText>

Build an argument to prove that the patient needed the procedure mentioned earlier.

Citing relevant parallels/precedents, give me ONLY the argument in the following format: Introduction, Procedures, Similar Cases, Conclusion.

Fig. 8: Baseline prompt template used with BioMedical-LLaMA.